

7.1 PERCENTAGE

You have often heard the sentences like “Sale Sale i.e Discount upto 50 percent”. Discount 10-60 percent and reduction 10% etc.

In our daily life, we apply percentage on many things like discounts, during shopping etc. To calculate percentage is an important skill during shopping.

Buy two
get one
FREE



The word percent is a short form of the Latin word “per centum”. Its meaning is “out of hundred”.

Recognize percentage as a fraction with denominator of 100

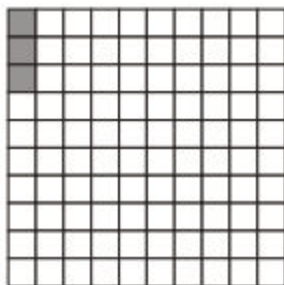
Look at this figure.

It has been divided into 100 equal parts.

3 out of 100 parts, are coloured.

i.e. $\frac{3}{100}$ of the figure is coloured.

We say that “3 percent” of the figure is coloured.



Percentage is denoted by the symbol %. So, we can say that 3% of the figure is coloured.

Any fraction with 100 as a denominator can be expressed as percentage.

Percentage is a special type of fraction which indicates parts out of hundred parts.

Convert a percentage to a fraction by expressing it as a fraction with denominator 100 and then simplify

We can convert a percentage to a fraction by expressing it as a fraction with denominator 100 and then simplify.

Consider the following examples.

Example 1: Convert 5% into fraction.

$$\begin{aligned}\text{Solution: } 5\% &= \frac{5}{100} \\ &= \frac{\cancel{5}}{\underset{20}{\cancel{100}}} = \frac{1}{20}\end{aligned}$$

Example 2: Convert 72% in fraction.

$$\begin{aligned}\text{Solution: } 72\% &= \frac{72}{100} \\ &= \frac{\overset{18}{\cancel{72}}}{\underset{25}{\cancel{100}}} = \frac{18}{25}\end{aligned}$$

Example 3: Convert $7\frac{1}{2}\%$ into fraction.

$$\begin{aligned}\text{Solution: } 7\frac{1}{2}\% &= \frac{15}{2}\% = \frac{15}{2} \times \frac{1}{100} \\ &= \frac{15}{200} \\ &= \frac{3}{40}\end{aligned}$$

Convert a fraction to a percentage by multiplying it with 100%

The fraction having denominators as 100 can easily be converted into percentage.

$$\text{For example: } \frac{5}{100} = 5\% \text{ and } \frac{97}{100} = 97\%$$

The fractions having denominators other than 100 are converted into percentage by two ways:

- (i) First convert the fraction into equivalent fraction having denominator as 100 and then express as percentage.
- (ii) Convert any fraction into percentage by multiplying and dividing with 100 or multiplying by 100%.

Example 1: Convert $\frac{1}{2}$ into percentage.

Solution: By converting into equivalent fraction with 100 denominator.

$$\frac{1}{2} = \frac{1 \times 50}{2 \times 50} = \frac{50}{100} = 50 \times \frac{1}{100} = 50\% \quad (\because \frac{1}{100} = 1\%)$$

By multiplying with 100%

$$\begin{aligned} \frac{1}{2} &= \frac{1}{2} \times \frac{100}{100} \\ &= \frac{100}{2} \times \frac{1}{100} = 50\% \quad (\because \frac{1}{100} = 1\%) \end{aligned}$$

Example 2: Convert $\frac{3}{5}$ into percentage.

Solution: By converting into equivalent fraction

$$\frac{3}{5} = \frac{3 \times 20}{5 \times 20} = \frac{60}{100} = 60\%$$

By multiplying with 100%

$$\begin{aligned} \frac{3}{5} &= \frac{3}{5} \times \frac{100}{100} \\ &= \frac{3 \times \overset{20}{\cancel{100}}}{\underset{5}{\cancel{100}}} \\ &= 60\% \quad (\because \frac{1}{100} = 1\%) \end{aligned}$$

Example 3: Convert $\frac{5}{8}$ into percentage.

Solution: $\frac{5}{8} = \frac{5}{8} \times \frac{100}{100}$
 $= (\frac{5}{8} \times \frac{25}{100}) \frac{1}{100} = \frac{125}{2} \%$

Convert a percentage to a decimal by expressing it as a fraction with denominator 100 and then as a decimal

Method of converting a percentage to a decimal is explained through following examples.

Example 1: Convert 3% into decimal.

Solution: $3\% = \frac{3}{100} = 0.03$

Example 2: Convert $\frac{5}{2} \%$ into decimal.

Solution: $\frac{5}{2} \% = 2.5\%$
 $= \frac{2.5}{100} = 0.025$

Example 3: Convert 135% into decimal.

Solution: $135\% = \frac{135}{100} = 1.35$

Convert a decimal to percentage by expressing it as a fraction with denominator 100 then as a percentage

Method of converting a decimal fraction into percentage is explained through following examples.

Example 1: Convert 0.5 into percentage.

Solution: $0.5 = \frac{5}{10} = \frac{5 \times 10}{10 \times 10} = \frac{50}{100} = 50\%$

Example 2: Convert 0.25 into percentage.

Solution: $0.25 = \frac{25}{100} = 25\%$

We can also convert decimal into percentage by multiply with 100%.

Example 3: Convert 0.125 into percentage.

Solution: $0.125 = 0.125 \times 100\%$
 $= \frac{125}{1000} \times 100\%$
 $= \frac{125}{10} \%$
 $= 12.5\%$

Finding percentage of a given quantity

In the following examples, we explain the method of finding the percent of a given quantity.

Example 1: Find 15% of 75.

Solution: $15\% \text{ of } 75 = \frac{15}{100} \times 75$
 $= \frac{45}{4}$
 $= 11\frac{1}{4}$

Example 2: Is 20% of 140 greater than 25% of 150.

Solution: $20\% \text{ of } 140 = 140 \times \frac{20}{100} = 28$
 $25\% \text{ of } 150 = 150 \times \frac{25}{100} = \frac{75}{2} = 37\frac{1}{2}$
As $28 < 37\frac{1}{2}$

Therefore, 20% of 140 is not greater than 25% of 150.

Example 3: Write the following fractions in ascending and descending order using percentage.

$$\frac{1}{2}, \frac{7}{5}, \frac{3}{4}$$

Solution: $\frac{1}{2} = \frac{1}{\cancel{2}^1} \times \frac{50}{100}\% = 50\%$

$$\frac{7}{5} = \frac{7}{\cancel{5}^1} \times \frac{20}{100}\% = 140\%$$

$$\frac{3}{4} = \frac{3}{\cancel{4}^1} \times \frac{25}{100}\% = 75\%$$

As $50\% < 75\% < 140\%$

So, Ascending order is:

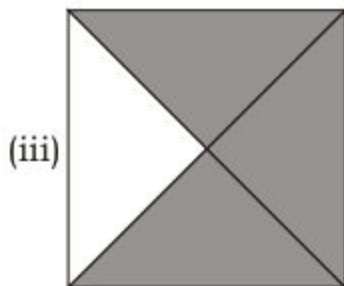
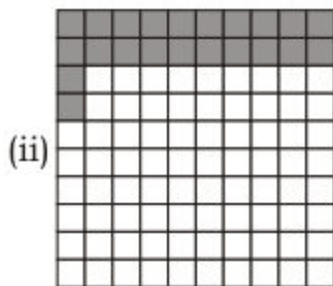
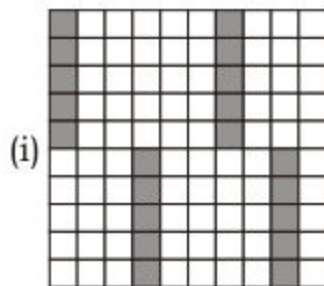
$$\frac{1}{2}, \frac{3}{4}, \frac{7}{5}$$

and Descending order is:

$$\frac{7}{5}, \frac{3}{4}, \frac{1}{2}$$

EXERCISE 7.1

1. What percent of the following figures is shaded.



2. Convert the following into fractions and decimal.

- (i) 25% (ii) 1% (iii) 31% (iv) 15%
(v) $8\frac{1}{2}\%$ (vi) 20.5% (vii) 175% (viii) 115%
(ix) 225% (x) $2\frac{1}{2}\%$

3. Convert the following fractions into percentage.

- (i) $\frac{1}{4}$ (ii) $\frac{7}{20}$ (iii) $\frac{8}{25}$ (iv) $\frac{5}{4}$

4. Write the following percentage into decimals.

- (i) 25% (ii) 31% (iii) 1% (iv) $2\frac{1}{2}\%$ (v) 6.5%

5. Convert the following decimal into percentage.

- (i) 0.025 (ii) 0.4 (iii) 0.85
(iv) 0.105 (v) 12.5 (vi) 125.5

6. What percentages are the following of 1 metre?

- (i) 70 cm (ii) 50 cm (iii) 75 cm (iv) 125 cm (v) 100 cm

7. The sale of the books on science on a certain store is 30%, write it in fraction.**8. Khalida obtained 96 marks out of 150. What percentage of marks has she obtained?****9. Find the following percentages.**

- | | |
|--------------------------|------------------------|
| (i) 10 % of 50 | (ii) 30 % of 70 |
| (iii) 25% of 250 | (iv) 50 % of 50 |
| (v) 75 % of 750 | (vi) 125% of 50 cm |
| (vii) 60 % of 300 litres | (viii) 20% of Rs 13.25 |
| (ix) 60.5 % of 8 hours | (x) 25% of 172 metres |
| (xi) 15 % of 210 | (xii) 75 % of 1011 |
| (xiii) 65% of 910 | (xiv) 35 % of 301 |

10. Write the following fractions in ascending and descending order using percentage.

$$\frac{2}{5}, \frac{1}{4}, \frac{3}{4}$$

Solve real life problems involving percentage

Percentages are used in daily life as explained in the following examples.

Example 1: Naheed has Rs 120. She spends 25% of the amount. How much money is left with her?

Solution: Total amount = Rs 120

Rate of expenditure = 25% of the total amount

$$\text{Amount spends} = \text{Rs } 120 \times \frac{25}{100} = 30$$

$$\text{Amount left with Naheed} = \text{Rs } 120 - 30 = 90$$

Example 2: 75% students out of 40 were present in a test. How many students were absent?

Solution: Total students = 40

percentage of present students = 75%

Number of students present = 75% of 40

$$= 40 \times \frac{75}{100} = 30 \text{ students}$$

$$\text{Number of students absent} = 40 - 30 = 10 \text{ students}$$

Second method.

As, percentage of the students present = 75%

So, percentage of the students absent = $100\% - 75\% = 25\%$

So, number of students who were absent = 25% of 40

$$= 40 \times \frac{25}{100} = 10 \text{ students}$$

Sometimes we need to find total quantity if percentage and given quantity (Percent of quantity) are given. In this case we use the rule

$$\text{Total quantity} = \text{Given quantity} \div \text{Percentage}$$

Example 3: Find the quantity if 75% of it is 123.

Solution: By using the formula

Required

$$\text{Quantity} = \text{Percent of the quantity} \div \text{Rate}$$

Here rate = 75% and percent of the quantity = 123

So, Required quantity = $123 \div 75\%$

$$= 123 \div \frac{75}{100}$$

$$= 123 \times \frac{100}{75}$$

$$= 164$$

Since, $\text{Percentage} = \frac{\text{Part of quantity}}{\text{Quantity}} \times 100\%$

Hence, the required quantity is 164.

Example 4: 25% of Hassan's salary is Rs 1250. Find his salary.

Solution: Here rate = 25%

Percent of salary = 1250 rupees

Now Salary = percent of salary $\div$ rate

$$= 1250 \div 25\%$$

$$= 1250 \div \frac{25}{100}$$

$$= 1250 \times \frac{100}{25}$$

$$= 5000 \text{ rupees}$$

Hence the salary of Hassan is Rs 5000

Example 5: In an examination Akram obtained 510 marks out of 850. Find the percentage of the marks obtained by him.

Solution: This question can be solved by the following formula.

$$\text{Percentage} = \frac{\text{Part of quantity}}{\text{Quantity}} \times 100\%$$

Here, Quantity = 850 marks

Part of quantity = 510 marks

$$\text{So, Percentage} = \frac{510}{850} \times 100\% = 60\%$$

EXERCISE 7.2

1. I obtained 60 marks out of 75 marks. What is the percentage of my marks?
2. Saima spent Rs 300 out of Rs 500. Find percentage of her expenditure.
3. There were 8500 voters in a village. 34% did not cast their votes. Find the number of voters who cast their votes.
4. The population of a village is 15000. If the population increases by 5% in a year, find the population after one year.
5. Rahila pays 5% of her salary in charity in a month. If she pays Rs 200, find her monthly salary.
6. The 18% of the distance between two cities is 36 km. Find the distance between two cities.
7. Naeem's monthly income is Rs 8000 and he spends Rs 6000 per month. Find the percentage of his expenditure.
8. On a rainy day 600 students out of 750 were present in a school. What percentage of the students were absent.

9. If one angle of a triangle is 90° . Find the other two angles if these angles are 30% and 70% of 90° respectively.
10. Area of square is 100 sq. cm. Find the area of a rectangle whose area is 10% more than the area of given square.
11. In a shop there is a double discount of 10% and 20% on an article having price of Rs 100. Calculate the total discount.

7.2 PROFIT, LOSS AND DISCOUNT

Define selling price and cost price

In every business goods are sold and purchased. Every business is established to earn profit but sometimes a loss to be borne.

The amount which is paid to purchase goods is called "**Cost price**" and the amount which is received by selling goods is called "**Selling price**".

Define profit, loss and discount

1. Profit Percentage and Loss Percentage

If in a business selling price of an item is more than cost price then there is a *profit* in such transaction. But if selling price of an item is less than cost price then there is a *loss* in such transaction.

Thus

Profit = Selling Price (SP) – Cost Price (CP)

Loss = Cost Price (CP) – Selling Price (SP)

Businessmen generally compare and calculate the profit and loss of goods in percent. For example if a shopkeeper purchases a juice for Rs 50 and sells for Rs 60. Similarly he purchases a milk pack for Rs 100 and sells for Rs 110. In both transactions, he earned Rs 10 as a profit. It seems that in above two cases he earned the same profit which is not fact. So, profit percent or loss percent are useful in such cases.

Now

$$\text{Profit percent} = \frac{\text{Profit}}{\text{CP}} \times 100\%$$

and

$$\text{Loss percent} = \frac{\text{Loss}}{\text{CP}} \times 100\%$$

$$\text{Profit Percent of juice} = \frac{10}{50} \times 100\% = 20\%$$

$$\text{Profit Percent of milk pack} = \frac{10}{100} \times 100\% = 10\%$$

This shows that the profit percent is not same in both cases.

Thus profit percent determines the transaction which gives more profit.

2. Discount

The reduction made on the list price or marked price of an article is called **discount** i.e. Net Selling price = Marked Price – Discount.

Remember: The discount percent is calculated only on the marked price and not on selling or cost price.

Example: The marked price of an article is Rs 400 and it is sold at a discount of 10%, find the net selling price of the article.

Solution: Discount = 10% of Rs 400

$$= \frac{10}{100} \times 400 = \frac{10}{100} \times 400 = \text{Rs } 40$$

So, Net Selling Price = Marked Price – Discount

$$= \text{Rs } 400 - \text{Rs } 40 = \text{Rs } 360$$

Solve real life problems involving profit, loss and discount

Example 1: A school bag is purchased for Rs 180 and is sold out for Rs 225. Find the profit percent.

Solution: Profit = selling price – cost price

$$= 225 - 180 = 45 \text{ rupees}$$

$$\text{Now Profit Percent} = \frac{\text{Profit}}{\text{CP}} \times 100\%$$

$$= \frac{45}{180} \times 100\% = \frac{450}{18} \%$$

$$= 25\%$$

So, Profit = 25%

Example 2: A fruit seller purchased 40 dozens bananas in all and their cost price is Rs 50 per dozen. He sells all the bananas at the rate of Rs 60 per dozen. Find his total profit and profit percent.

Solution: Bananas = 40 dozen
Cost price per dozen = 50 rupees
Total cost price = $50 \times 40 = 2000$ rupees
Selling price per dozen = 60 rupees
Total selling price = $60 \times 40 = 2400$ rupees
Profit = selling price – cost price
= $2400 - 2000 = 400$ rupees

$$\begin{aligned}\text{Profit Percent} &= \frac{\text{Profit}}{\text{CP}} \times 100\% \\ &= \frac{400}{2000} \times 100\% \\ &= \frac{1}{5} \times 100\% \\ &= 20\%\end{aligned}$$

Example 3: Saima purchased a mobile set for Rs 15000. After some time she sold it for Rs 12000. Find her loss percent.

Solution: Here CP = 15000 rupees
SP = 12000 rupees
Since $SP < CP$, so there is loss
Now Loss = CP – SP
= $15000 - 12000 = 3000$ rupees

$$\begin{aligned}\text{So, Loss Percent} &= \frac{\text{Loss}}{\text{CP}} \times 100\% \\ &= \frac{3000}{15000} \times 100\% \\ &= \frac{1}{5} \times 100\% \\ &= 20\%\end{aligned}$$

Example 4: Bismah purchased a T.V for Rs 20000 and sold it for Rs 18000 to purchase a new model T.V. Find her profit or loss percent.

Solution: Here CP = 20000 rupees

$$SP = 18000 \text{ rupees}$$

Since $SP < CP$, so there is loss

$$\text{Now, Loss} = CP - SP$$

$$= 20000 - 18000 = 2000 \text{ rupees}$$

$$\text{Loss Percent} = \frac{\text{Loss}}{CP} \times 100\%$$

$$= \frac{2000}{20000} \times 100\%$$

$$= \frac{1}{10} \times 100\%$$

$$= 10\%$$

Example 5: A trader marked his goods at 30% above the cost price. If he allows his customer 20% discount, find his profit percent.

Solution: Let the cost price be Rs 100.

$$\text{Then marked price} = \text{Rs } (100 + 30) = \text{Rs } 130$$

$$\text{Discount} = 20\% \text{ on marked price}$$

$$\frac{20}{100} \times 130 = \text{Rs } 26$$

$$\text{So, Net Selling Price} = \text{Marked Price} - \text{Discount}$$

$$= 130 - 26 = \text{Rs } 104$$

$$\text{Thus Profit} = \text{Selling Price} - \text{Cost Price}$$

$$= 104 - 100 = \text{Rs } 4$$

Hence, the profit is 4%.

EXERCISE 7.3

- 1. Find the profit or loss percent in the following.**
 - (i) Eggs are purchased at Rs 120 per dozen and are sold out at Rs 130 per dozen.
 - (ii) A shopkeeper purchases 25 breads at Rs 10 per bread. He sells breads at Rs 13 per bread as 7 breads are spoiled.
 - (iii) A man purchases pencils at Rs 60 per dozen and sells at Rs 70 per dozen.
 - (iv) A man purchases 10 dozen eggs at the rate of Rs 120 per dozen. 3 dozen eggs are broken. Rest of the eggs are sold at the rate of Rs 130 per dozen.
 - (v) A milkman purchases milk at the rate of Rs 75.50 per litre and sells at the rate of Rs 80 per litre.
- 2.** A book seller purchases 20 books of Mathematics at Rs. 120 per book and sells at Rs 125 per book. Find his profit and profit percent.
- 3.** A grocer purchases 10 kilograms of rice at Rs 120 per kilogram and sells at Rs 125 per kilogram. Find profit percent.
- 4.** A shopkeeper bears 15% loss on the sale of a jacket. If the cost of the jacket is Rs 1000, find his loss.
- 5.** Umaima bought a dinner set for Rs 52000 at 20% discount. Find the actual price of the dinner set.
- 6.** The marked price of ladies purse is Rs 1500 and it is sold at discount of 15%. Find net selling price of the purse.

REVIEW EXERCISE 7

1. Karim bought 50 dozen of pencils at Rs 60 per dozen. He sold them at Rs 13 per pencil. Find his profit or loss percent.
2. A television (T.V) manufacturing company declares that a colour T.V is now available for Rs 13050. Its previous price was Rs 17400. Find the percentage reduction in the price of colour T.V offered by the company.
3. Imran obtained 548 marks out of 800 and his sister obtained 459 out of 600. Whose performance is better?
4. A fruit seller purchases 50 dozen bananas at the rate of Rs 60 per dozen. 10 dozen bananas are spoiled. Rest of the bananas are sold at the rate of Rs 70 per dozen. Find his gain or loss percent.
5. A car was sold at price of Rs 875,000 with a discount of $2\frac{1}{2}\%$. What is actual cost of the car?
6. **Which is greater?**
 - (i) 0.045 or 5% (ii) 0.64 or 62% (iii) 0.8 or 8%
 - (iv) 0.15 or 19% (v) $\frac{1}{2}$ or 40% (vi) $\frac{1}{5}$ or 25%
 - (vii) $\frac{3}{4}$ or 70% (viii) $1\frac{1}{4}$ or 130% (ix) $\frac{1}{10}$ or 9%
7. **(a) Fill in the blanks:**
 - (i) $\frac{3}{4} = \text{————}\%$ (ii) $\frac{5}{8} = \text{————}\%$
 - (iii) $30\% = \text{————}$ (fraction)
 - (iv) $0.125 = \text{————}\%$
 - (v) $38\% = \text{————}$ (decimal)
 - (vi) $36\% \text{ of } 120 = \text{————}$ (vii) $\text{S.P} + \text{Loss} = \text{————}$
 - (viii) $\text{S.P} - \text{C.P} = \text{————}$

(b) Write true (T) or false (F) for the following.

(i) If $S.P < C.P$ so, there is gain.

(ii) If $C.P > S.P$ so, there is loss.

(iii) $\text{Loss} = C.P - S.P$

(iv) $S.P + \text{gain} = C.P$

(v) $\text{Loss \%} = \frac{\text{Loss}}{S.P} \times 100\%$

(vi) $\text{Profit} = S.P - C.P$

(vii) $\frac{\text{Profit} \times 100\%}{C.P} = \text{Profit \%}$

SUMMARY

- The word “**percent**” means “**out of hundred**”. Percentage is denoted by the symbol %
- Percentage can be converted into **common** or **decimal fraction**.
 - (a) To convert a percent into a common fraction, we multiply the number indicating the percent by $\frac{1}{100}$ and simplify the resulting fraction.
 - (b) To convert a percent into a decimal, we remove the % symbol and shift the decimal point two places to the left.
- Fraction or a decimal fraction can be converted into percentage.
 - (a) To convert a decimal into a percent, we shift the decimal point two places to the right and insert percent symbol (%).
 - (b) (i) The fractions having their **denominators** as 100 can easily be converted into percentage.

(ii) The fractions having their denominators other than 100 are converted into percentages in the following ways:

- ★ First convert the fraction into equivalent fraction having denominator as 100 and then convert it into percentage.
- ★ First convert the fraction into decimal and then convert it into percentage.
- ★ Multiply the given fraction by 100%.

- Percent of a quantity = Quantity $\times$ rate
- Quantity = Percentage of Quantity $\div$ rate
- Percentage = $\frac{\text{Part of Quantity}}{\text{Whole Quantity}} \times 100\%$
- When S.P > C.P, then Profit = S.P – C.P
- When S.P < C.P, then Loss = C.P – S.P
- Profit % = $\frac{\text{Profit}}{\text{C.P}} \times 100\%$ and Loss % = $\frac{\text{Loss}}{\text{C.P}} \times 100\%$